LAB 2: PL/SQL part 1.

PL/SQL fundamentals

1. *Write a PL/SQL block to show an invalid case-insensitive reference to a quoted and without quoted user-defined identifier.*

After running these commands:

```
DECLARE
  "MyVar" NUMBER := 10;
BEGIN
  DBMS_OUTPUT.PUT_LINE("MYVAR");
END;
/
```

an error appears as expected.

After running these commands:

```
DECLARE
  myvar NUMBER := 10;
BEGIN
  DBMS_OUTPUT.PUT_LINE(MYVAR);
END;
/
```

the message I receive is:
PL/SQL procedure successfully completed.

Quoted identifiers in Oracle are case-sensitive.

Unquoted identifiers are case-insensitive, so myvar, MYVAR, and MyVar all refer to the same variable.

2. *Write a PL/SQL block to show the result to neglect double quotation marks in reserved word identifier.*

After running these commands:

```
DECLARE
  "SELECT" VARCHAR2(20) := 'Hello';
BEGIN
  DBMS_OUTPUT.PUT_LINE("SELECT");
END;
/
```

the message I receive is:
PL/SQL procedure successfully

After running these commands:

```
DECLARE
  "SELECT" VARCHAR2(20) := 'Hello';
BEGIN
  DBMS_OUTPUT.PUT_LINE(SELECT);
END;
/
```

a compilation error appears.

The identifier "SELECT" is valid only because it is enclosed in double quotes. When the quotes are omitted, Oracle interprets SELECT as the SQL reserved word rather than the declared variable, causing a compilation error.

3. *Write PL/SQL blocks to show the declaration of 5 variables. The data type of 3 variables must be the same (imported) as data type of the first columns in table hr.employee.*

First I created a teble employees:

```
CREATE TABLE employees (
  employee_id NUMBER,
  first_name VARCHAR2(40),
  last_name VARCHAR2(40),
  salary NUMBER,
  hire_date DATE,
  manager_id NUMBER,
  department NUMBER
);
```

declaration of 5 variables :

```
DECLARE
  v_employee_id employees.employee_id%TYPE;
  v_first_name  employees.first_name%TYPE;
  v_last_name   employees.last_name%TYPE;

  v_salary NUMBER;
  v_message VARCHAR2(100);
BEGIN
  NULL;
END;
/
```

4. *Create table temp containing 4 columns:*
   a. ID number
   b. FirstName varchar2(40)
   c. LastName varchar2(40)
   d. Salary number

First let's add data:

```
INSERT INTO employees VALUES
(100,'Steven','King',24000,DATE '2003-06-17',NULL,90);
INSERT INTO employees VALUES
(101,'Neena','Kochhar',17000,DATE '2005-09-21',100,90);
INSERT INTO employees VALUES
(102,'Lex','De Haan',17000,DATE '2001-01-13',100,90);
INSERT INTO employees VALUES
(103,'Alexander','Hunold',9000,DATE '2006-01-03',102,50);
COMMIT;
```

Then create temp:

```
CREATE TABLE temp (
  id NUMBER,
  firstname VARCHAR2(40),
  lastname VARCHAR2(40),
  salary NUMBER
);
```

Adding data to it:

```
INSERT INTO temp VALUES (100, 'Steven', 'King', NULL);
INSERT INTO temp VALUES (101, 'Neena', 'Kochhar', NULL);
INSERT INTO temp VALUES (102, 'Lex', 'De Haan', NULL);
INSERT INTO temp VALUES (103, 'Alexander', 'Hunold', NULL);
COMMIT;
```

PL/SQL block to copy salaries:

```
BEGIN
  FOR emp_rec IN (
    SELECT employee_id, salary
    FROM employees
  ) LOOP
    UPDATE temp
    SET salary = emp_rec.salary
    WHERE id = emp_rec.employee_id;
  END LOOP;
  COMMIT;
END;
/
```

And to verify:
```
SELECT *
FROM temp
ORDER BY id;
```

This image shows the output as expected

ID	FIRSTNAME	LASTNAME	SALARY
100	Steven	King	24000
101	Neena	Kochhar	17000
102	Lex	De Haan	17000
103	Alexander	Hunold	9000

PL/SQL control statement

1. *Write a PL/SQL program to check whether a number is even or odd.*

```
DECLARE
  v_number NUMBER := 18;
BEGIN
  IF MOD(v_number, 2) = 0 THEN
    DBMS_OUTPUT.PUT_LINE(v_number || ' is even');
  ELSE
    DBMS_OUTPUT.PUT_LINE(v_number || ' is odd');
  END IF;
END;
/
```

The result: 18 is even

2. *Write a PL/SQL program to check whether a date from column hr.employees.hirdate falls on Monday or Friday.*

```
DECLARE
  v_day VARCHAR2(20);
BEGIN
  SELECT TO_CHAR(hire_date, 'DAY')
  INTO v_day
  FROM employees
  WHERE employee_id = 100;
  IF TRIM(v_day) = 'MONDAY' THEN
    DBMS_OUTPUT.PUT_LINE('Monday');
  ELSIF TRIM(v_day) = 'FRIDAY' THEN
    DBMS_OUTPUT.PUT_LINE('Friday');
  ELSE
    DBMS_OUTPUT.PUT_LINE('Other day');
  END IF;
END;
/
```

The result: Other day

3. *Write a PL/SQL program to display all rows from table hr.employees (replace manager ID with manager name).*

```
BEGIN
  FOR emp_rec IN (
    SELECT e.employee_id,
         e.first_name,
         e.last_name,
         NVL(m.first_name || ' ' || m.last_name,
           'No Manager') AS manager_name
    FROM employees e
    LEFT JOIN employees m
      ON e.manager_id = m.employee_id
    ORDER BY e.employee_id
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE(
      emp_rec.employee_id || ' ' ||
      emp_rec.first_name || ' ' ||
      emp_rec.last_name || ' Manager: ' ||
      emp_rec.manager_name
    );
  END LOOP;
END;
/
```

The result:
100 Steven King Manager:
101 Neena Kochhar Manager: Steven King
102 Lex De Haan Manager: Steven King
103 Alexander Hunold Manager: Lex De Haan

4. *Write a PL/SQL program to count number of employees in all department. (HR schema)*

```
SELECT department,
     COUNT(*) AS employee_count
FROM employees
GROUP BY department
ORDER BY department;
```

The result:

```
DEPARTMENT EMPLOYEE_COUNT
---------- --------------
        50              1
        90              3
```

5. *Write a PL/SQL program to check whether a salary of a given employess is smaller or greater than average salary in department to which employees belongs.*

```
SET SERVEROUTPUT ON;
DECLARE
  v_employee_id NUMBER := 103;

  v_salary NUMBER;
  v_department NUMBER;
  v_avg_salary NUMBER;

BEGIN
  SELECT salary, department
  INTO v_salary, v_department
  FROM employees
  WHERE employee_id = v_employee_id;

  SELECT AVG(salary)
  INTO v_avg_salary
  FROM employees
  WHERE department = v_department;

  IF v_salary > v_avg_salary THEN
    DBMS_OUTPUT.PUT_LINE(
      'Salary is greater than department average'
    );
  ELSIF v_salary < v_avg_salary THEN
    DBMS_OUTPUT.PUT_LINE(
      'Salary is smaller than department average'
    );
  ELSE
    DBMS_OUTPUT.PUT_LINE(
      'Salary equals department average'
    );
  END IF;
END;
/
```

For employee with id number 103 the result is: Salary equals department average